



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. (General) Degree in Applied Sciences
First Year – Semester I Examination – June/ July 2018

FDN 1204 – BASIC MATHEMATICS

Time: Two (02) hours

Answer all questions.

1. (a) Solve each of the following inequalities and express the solution on a real number line.

(i) $2(3x+4) > 1-x$

(ii) $|x-4| \leq 3$

(20 marks)

- b) Find the solutions of each of the following equations.

(i) $x^2 + 7x + 6 = 0$

(ii) $2x^2 + x - 10 = 0$

(40 marks)

- c) Sketch the graph of the equation, $y = x^2 - 2x - 3$ in a Cartesian coordinate system and identify the x intercepts and the axis of symmetry of the graph.

(40 marks)

2. a) Find the exact value of each of the followings.

(i) $\cos(105^\circ)$

(ii) $\sin\left(-\frac{\pi}{12}\right)$

(20 marks)

b) Verify the following Trigonometric identities.

(i) $5 \sin^2 x + 4 \cos^2 x = \sin^2 x + 4$

(ii) $\frac{\sin x + \sin 3x}{\cos x + \cos 3x} = \tan 2x$

(40 marks)

c) Find the limits of the followings.

(i) $\lim_{x \rightarrow 3} \frac{x^2 - 7x + 12}{x - 3}$

(ii) $\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2 - x - 2}$

(40 marks)

3. a) Let $y = e^{-x}(\cos 2x + \sin 2x)$

(i) Show that, $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 0$.

(ii) Find $\left(\frac{d^3 y}{dx^3} \right)_{x=0}$

(60 marks)

b) Find each of the following integrations.

(i) $\int (x^{\frac{2}{3}} + 2x + 3) dx$

(ii) $\int \frac{(x^2 + 5x + 1)}{x^2} dx$

(40 marks)

4. a) Consider the matrices, $A = \begin{pmatrix} 2 & -1 \\ 5 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 & -2 \\ -1 & 3 & 5 \end{pmatrix}$ and $C = \begin{pmatrix} -7 & -2 \\ 1 & 3 \end{pmatrix}$ and

find the followings.

(i) $A + C$

(ii) AB

(iii) Transpose of matrix B, (B^T)

(iv) Inverse of matrix A, (A^{-1})

(50 marks)

- b) Let $\underline{u} = (-3, 4)$ and $\underline{v} = (-1, 5)$ are two vectors and hence find the followings.

(i) $3\underline{u} - 2\underline{v}$

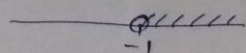
(ii) Magnitude of vector $\underline{u}$

(iii) Unit vector of vector $\underline{u}$

(50 marks)

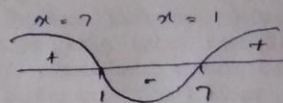
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$$\begin{aligned} \textcircled{1} \text{ a) } 2(3x+4) &> 1-x \\ 6x+8 &> 1-x \\ 7x &> -7 \\ x &> -1 \end{aligned}$$



$$\underline{\underline{x \in (-1, \infty)}}$$

$$\begin{aligned} |x-4| &\leq 3 \\ (x-4)^2 &\leq 3^2 \\ x^2-8x+16 &\leq 9 \\ x^2-8x+7 &\leq 0 \\ (x-7)(x-1) &\leq 0 \end{aligned}$$



$$\underline{\underline{x \in [1, 7]}}$$

$$\begin{aligned} \text{b) } x^2+7x+6 &= 0 \\ x^2+6x+x+6 &= 0 \\ x(x+6)+1(x+6) &= 0 \\ (x+1)(x+6) &= 0 \\ x+1=0 &\quad \text{or} \quad x+6=0 \\ \underline{\underline{x=-1}} &\quad \underline{\underline{x=-6}} \end{aligned}$$

$$\begin{aligned} 2x^2+x-10 &= 0 \\ 2x^2+5x-4x-10 &= 0 \\ x(2x+5)-2(2x+5) &= 0 \\ (2x+5)(x-2) &= 0 \\ 2x+5=0 &\quad \text{or} \quad x-2=0 \\ \underline{\underline{x=-\frac{5}{2}}} &\quad \underline{\underline{x=2}} \end{aligned}$$

$$\begin{aligned} \text{c) } y &= x^2-2x-3 \\ x=0 \text{ විට} \\ y &= -3 \\ (0, -3) \end{aligned}$$

$$\begin{aligned} y &= 0 \text{ විට} \\ 0 &= x^2-2x-3 \\ 0 &= (x-3)(x+1) \\ x &= 3 \quad x = -1 \\ (3, 0) \quad (-1, 0) \end{aligned}$$

$a > 0$ නිසා උඩට.

$$\begin{aligned} y &= (x-1)^2-3-1 \\ &= (x-1)^2-4 \end{aligned}$$

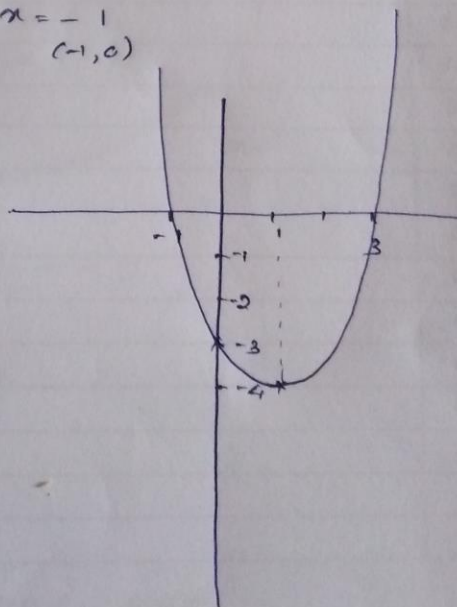
$$\text{ඉ } x=1 \text{ විට } y=-4$$

නමුත් x වල මැද y ගත

$$\text{ඉ } x = \frac{3+(-1)}{2} = 1$$

$$\begin{aligned} \text{එකල } y &= 1-2-3 \\ &= -4 \end{aligned}$$

$$\text{නමුත් මධ්‍යම } = (1, -4)$$



Q) a) $\cos(105^\circ) = \cos(180 - 75)$
 $= -\cos 75$
 $= -\cos(45+30)$

S	A
T	C

$$= -[\cos 45 \cdot \cos 30 - \sin 45 \cdot \sin 30]$$

$$= -\left[\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}\right]$$

$$= -\frac{(\sqrt{3}-1)}{2\sqrt{2}} = \frac{-\sqrt{3}+1}{2\sqrt{2}}$$

Q) $\cos\left(\frac{\pi}{12}\right) \sin\left(-\frac{\pi}{12}\right) = -\sin\left(\frac{\pi}{12}\right)$
 $= -\sin\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$

$$= -\left[\sin \frac{\pi}{4} \cdot \cos \frac{\pi}{6} - \sin \frac{\pi}{6} \cdot \cos \frac{\pi}{4}\right]$$

$$= -\left[\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{\sqrt{2}}\right]$$

$$= -\frac{(\sqrt{3}-1)}{2\sqrt{2}} = \frac{1-\sqrt{3}}{2\sqrt{2}}$$

b) i) $5 \sin^2 x + 4 \cos^2 x = \sin^2 x + 4$
 ~~$5 \sin^2 x + 4(1 - \sin^2 x) = \sin^2 x + 4$~~
 ~~$\sin^2 x$~~
 L.H.S
 $5 \sin^2 x + 4 \cos^2 x$
 $5 \sin^2 x + 4(1 - \sin^2 x)$
 $\underline{\underline{\sin^2 x + 4}} \quad \text{L.H.S} = \text{R.H.S}$

ii) $\frac{\sin x + \sin 3x}{\cos x + \cos 3x} = \tan 2x$

L.H.S.
 $\frac{2 \sin 2x \cdot \cos(-x)}{2 \cos 2x \cdot \cos(-x)}$
 $\underline{\underline{\tan 2x}} \quad \text{L.H.S} = \text{R.H.S}$

$$c) \quad x \lim_{x \rightarrow 3} \frac{x^2 - 7x + 12}{(x-3)}$$

$$x \lim_{x \rightarrow 3} \frac{(x-4)(x-3)}{(x-3)}$$

$$x \lim_{x \rightarrow 3} (x-4)$$

$$3-4$$

$$\underline{\underline{-1}}$$

$$n) \quad x \lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2 - x - 2}$$

$$x \lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)(x+1)}$$

$$x \lim_{x \rightarrow 2} \frac{\sin(x-2)}{x-2} \times \frac{1}{x \lim_{x \rightarrow 2} (x+1)}$$

$$x \lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)} \times \frac{1}{x \lim_{x \rightarrow 2} (x+1)}$$

$$1 \times \frac{1}{2+1}$$

$$\underline{\underline{\frac{1}{3}}}$$

$$3) \quad y = e^{-x} (\cos 2x + \sin 2x) \rightarrow (1) \quad y(x=0) = 1 \times (1+0) = \underline{\underline{1}}$$

$$\frac{dy}{dx} = e^{-x} (-\sin 2x \times 2 + \cos 2x \times 2) + (\cos 2x + \sin 2x) e^{-x}$$

$$\frac{dy}{dx} = 2e^{-x} (\cos 2x - \sin 2x) - y \rightarrow (2) \quad \left(\frac{dy}{dx}\right)_{x=0} = 2 \times 1 (1-0) - 1 = 2-1 = \underline{\underline{1}}$$

এই মত x সিস্টেম অপারেটর অভ্যাস.

$$\frac{d^2y}{dx^2} = 2e^{-x} (-\sin 2x \times 2 - \cos 2x \times 2) - \frac{dy}{dx} + 2(\cos 2x - \sin 2x) e^{-x} - 1 - \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = -4e^{-x} (\cos 2x + \sin 2x) - 2e^{-x} (\cos 2x - \sin 2x) - \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = -4y + \left(-\frac{dy}{dx} - y\right) - \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = -5y - 2\frac{dy}{dx} \rightarrow (3) \quad \left(\frac{d^2y}{dx^2}\right)_{x=0} = -5 \times 1 - 2 \times 1 = \underline{\underline{-7}}$$

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$$

5) a) නමුත් x විෂයේ අවකලනය,

$$\frac{d^3y}{dx^3} = -5 \frac{dy}{dx} - 2 \frac{d^2y}{dx^2}$$

$$\begin{aligned} \left(\frac{d^3y}{dx^3} \right)_{x=0} &= -5 \left(\frac{dy}{dx} \right)_{x=0} - 2 \left(\frac{d^2y}{dx^2} \right)_{x=0} \\ &= -5 \times 1 - 2 \times -7 \\ &= -5 + 14 \\ &= \underline{\underline{9}} \end{aligned}$$

b) $I = \int x^{\frac{2}{3}} + 2x + 3 \, dx$

$$= \int x^{\frac{2}{3}} \, dx + 2 \int x \, dx + 3 \int dx$$

$$= \frac{x^{\frac{5}{3}}}{\frac{5}{3}} + \frac{2x^2}{2} + 3x + K$$

$$= \underline{\underline{\frac{3}{5} x^{\frac{5}{3}} + x^2 + 3x + K}}$$

K - නිශ්චය නොවේ.

$$I = \int \frac{x^2 + 5x + 1}{x^2} \, dx$$

$$= \int 1 + \frac{5}{x} + \frac{1}{x^2} \, dx$$

$$= \int 1 + (5x^{-1}) + x^{-2} \, dx$$

$$= \int dx + 5 \int x^{-1} \, dx + \int x^{-2} \, dx$$

$$= \int dx + \int x^{-2} \, dx + 5 \int \frac{1}{x} \, dx$$

$$= x + \frac{x^{-1}}{-1} + 5 \ln|x| + C$$

$$= \underline{\underline{x - \frac{1}{x} + 5 \ln|x| + C}}$$

C - නිශ්චය නොවේ

$$\textcircled{1} A = \begin{bmatrix} 2 & -1 \\ 5 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & -2 \\ -1 & 3 & 5 \end{bmatrix} \quad C = \begin{bmatrix} -7 & -2 \\ 1 & 3 \end{bmatrix}$$

$$\textcircled{1} A + C = \begin{bmatrix} 2 & -1 \\ 5 & 4 \end{bmatrix} + \begin{bmatrix} -7 & -2 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & -3 \\ 6 & 7 \end{bmatrix}$$

$$\textcircled{2} AB = \begin{bmatrix} 2 & -1 \\ 5 & 4 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 & 0 & -2 \\ -1 & 3 & 5 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 2 \times 1 + (-1)(-1) & 2 \times 0 + (-1) \times 3 & 2 \times -2 + 5 \times -1 \\ 5 \times 1 + 4 \times -1 & 5 \times 0 + 4 \times 3 & 5 \times -2 + 4 \times 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2 + 1 & 0 - 3 & -4 - 5 \\ 5 - 4 & 0 + 12 & -10 + 20 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -3 & -9 \\ 1 & 12 & 10 \end{bmatrix}$$

$$\textcircled{3} B^T = \begin{bmatrix} 1 & -1 \\ 0 & 3 \\ -2 & 5 \end{bmatrix}_{3 \times 2}$$

$$A^{-1} = \frac{1}{2 \times 4 - (5 \times -1)} \begin{bmatrix} 4 & 1 \\ -5 & 2 \end{bmatrix}$$

$$= \frac{1}{13} \begin{bmatrix} 4 & 1 \\ -5 & 2 \end{bmatrix}$$

$$b) \underline{u} = (-3, 4)$$

$$\underline{u} = -3\underline{i} + 4\underline{j}$$

$$\underline{v} = (-1, 5)$$

$$\underline{v} = -\underline{i} + 5\underline{j}$$

$$\begin{aligned} 1) \quad 3\underline{u} - 2\underline{v} &= 3(-3\underline{i} + 4\underline{j}) - 2(-\underline{i} + 5\underline{j}) \\ &= -9\underline{i} + 12\underline{j} + 2\underline{i} - 10\underline{j} \\ &= -7\underline{i} + 2\underline{j} \end{aligned}$$

$$\begin{aligned} 2) \quad |\underline{u}| &= \sqrt{(-3)^2 + (4)^2} \\ &= \sqrt{9 + 16} \\ &= \underline{\underline{5}} \end{aligned}$$

3)

$$3) \quad \text{Unit vector} = \frac{\underline{u}}{|\underline{u}|} = \frac{-3\underline{i} + 4\underline{j}}{5} = \underline{\underline{\frac{-3}{5}\underline{i} + \frac{4}{5}\underline{j}}}$$

